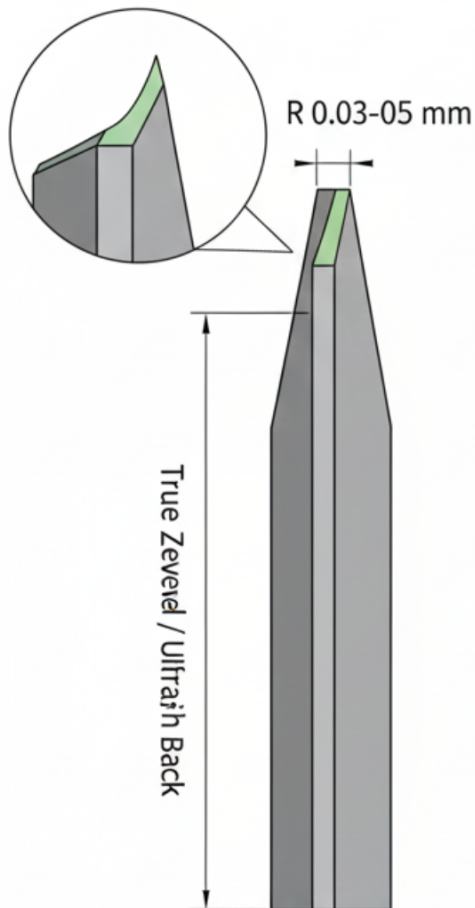


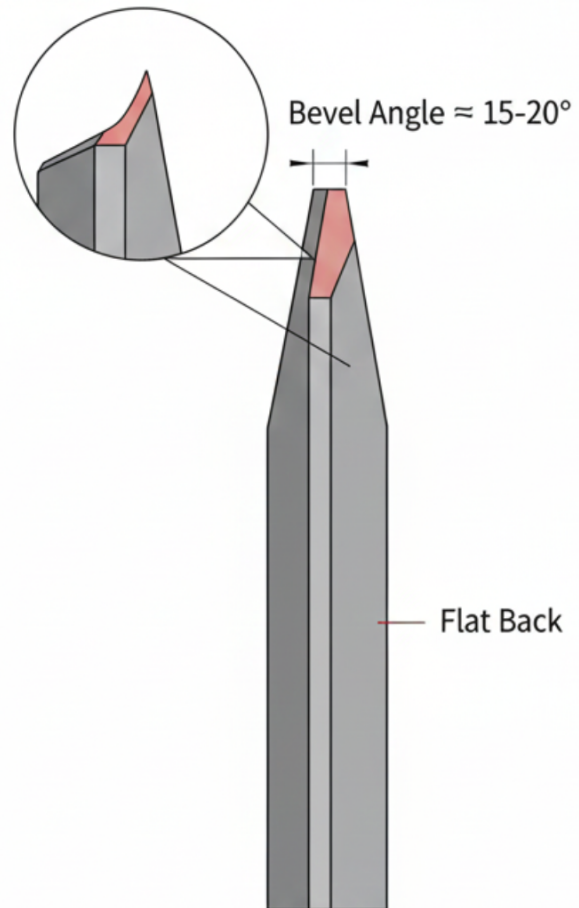
LEFT: Micro Radius
(Stabilized Flush Edge)

Micro Radius: Prevents Chipping, Distributes Stress. No Burr Formation.



RIGHT: Micro Bevel
(Classic Beveled Edge)

Micro Bevel: Common, Adds Strength, But Not Truly "Flush".



Source: JUWEX TX-5 Engineering.

“This image is only to clarify that we do not want micro bevel. Required edge geometry is micro radius, as visually shown.”

OFFICIAL PRODUCTION MANUAL: JUWEX TX-5

Document Ref: PRO-MAN-TX5-2026

Subject: High-Precision Ultra-Flush
Industrial Cutters

1. MATERIAL SPECIFICATIONS

- **Steel Grade:** High-Performance 65CrMo (Chrome-Molybdenum Alloy).
- **Forging:** Precision Drop Forged for superior grain flow and structural density.

2. HEAT TREATMENT & HARDNESS STANDARDS

- **Cutting Edge Hardness:** 59-60 HRC (Target: 60 HRC).
- **Body Hardness:** 54-55 HRC (To

ensure shock absorption and flexibility).

- **Process:** Computer-controlled Induction Hardening with a shallow depth (0.3mm–0.5mm) to maintain a soft core.

3. EDGE GEOMETRY & GRINDING (CRITICAL)

- **Cut Profile: Ultra-Flush / No-Bevel.**
The exterior face of the blade must be ground 100% flat.
- **The Edge Apex (Micro-Radius):** A radius of 0.03 mm – 0.05 mm must be applied to the cutting tip post-sharpening.
- *Constraint:* No secondary bevel is allowed. The radius must be microscopic and uniform.

- **Honing:** Final edges must be polished to **8000+ Grit** mirror finish to minimize frictional heat.

4. OPTICAL LASER ALIGNMENT (OLA)

- **Alignment Requirement:** 100% of units must be verified via laser scanning.
- **Tolerance:** Blade parallelism deviation < 0.02 mm.
- **Zero-Gap Standard:** When closed, the blades must exhibit **zero light leakage** under a high-intensity light source.

5. ESD PROTECTION & ERGONOMICS

- **Handles:** ESD-Safe (Antistatic) PVC/

TPE double-dipped grips.

- **Surface Resistivity:** $10^{\{6\}}$ to $10^{\{9\}}$ ohms (Compliant with electronics industry standards).
- **Spring System:** Internal hidden spring mechanism for constant tension and durability.

6. SURFACE FINISH & BRANDING

- **Coating:** High-gloss **Transparent Epoxy-Resin** (anti-corrosion).
- **Blade Clearance:** The cutting edge must remain free of epoxy residue to ensure surgical precision.
- **Marking:** High-resolution laser etching: "JUWEX TX-5 | 60 HRC | ESD-Safe | Industrial Precision".

7. CUTTING CAPACITY LIMITS

(OPERATIONAL)

Soft Wire (Cu, Au, Ag)

1.6 mm

Medium Wire (Iron)

0.60 mm

Hard Wire

Not Recommended